

Urolithiasis



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KEYWORDS

- Lower urinary tract • Urolithiasis • Urinary calculi • Struvite • Calcium oxalate
- Purine • Cystine

KEY POINTS

- Urolithiasis occurs commonly in dogs and cats, and most uroliths occur in the lower urinary tract.
- More than 80% to 90% of lower urinary tract uroliths are struvite or calcium oxalate.
- Some uroliths, such as struvite, cysteine, and urate, are amenable to medical dissolution, whereas others, such as calcium oxalate, are not.

INTRODUCTION

Formation of uroliths is not a disease but rather a complication of several disorders. Some disorders can be identified and corrected (such as infection-induced struvite urolith formation), some can be identified but not corrected (such as hyperuricosuria, which occurs in Dalmatians that excrete high levels of uric acid, which forms ammonium urate uroliths), whereas for others the underlying etiopathogenesis is not known (such as calcium oxalate urolith formation in miniature schnauzers). A common denominator of these disorders is that they can from time to time create oversaturation of urine with 1 or more crystal precursors, resulting in formation of crystals. To develop rational and effective approaches to treatment, abnormalities that promote urolith formation must be identified, with the goal of eliminating or modifying them. It is important, therefore, to understand several basic concepts associated with urolithiasis and the factors that promote urolith formation that may be modified with medical treatment, including the state of urinary saturation, modifiers of crystal formation, presence of multiple crystal types, and presence of bacterial infection, urinary obstruction, or foreign compounds.¹ Urolith formation, dissolution, and prevention involve complex physical processes. Major factors include (1) supersaturation resulting in crystal formation, (2) effects of inhibitors of crystallization and inhibitors of crystal aggregation and growth, (3) crystalloid complexors, (4) effects of promoters of crystal aggregation and growth, and (5) effects of noncrystalline matrix.¹

The authors have nothing to disclose.

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DIAGNOSIS OF UROLITHS

Imaging is the most definitive diagnostic tool for detection of uroliths. Abdominal radiography is generally the first diagnostic imaging modality used to detect radiopaque uroliths (**Fig. 1**). Ultrasonography (**Fig. 2**) or double contrast cystography (**Fig. 3**) can be used to detect uroliths, including those that are radiolucent.² These abdominal imaging techniques are used to verify the presence of uroliths and location, number, size, shape, and density.

In patients with suspected urinary tract disorders, urinalysis is an important part of diagnostic evaluation. Crystalluria can be an important finding (**Fig. 4**). Crystals do not confirm the presence of uroliths, but they do suggest crystalline oversaturation, and some patients may have active urocystoliths present but not have crystalluria.³ Temperature change caused by elapsed time between urine collection and urinalysis can cause crystals to form in urine, resulting in a false-positive crystalluria.⁴ Therefore, in patients with suspected urolithiasis, freshly collected urine should be evaluated.⁵

Urine specific gravity and urine pH can help assess the chemical environment of urine. The chemical environment of the urine determines urolith formation and can suggest which type of urolith is present. A high urine specific gravity suggests an increase in concentration of urolithic precursors.⁶ Calcium oxalate, purines, and cystine uroliths form typically in urine with a pH less than 7.0, whereas struvite calculi form typically in urine with a pH greater than 7.0.⁵

Urine culture and sensitivity testing are indicated because urinary tract infections may occur secondarily in patients with urolithiasis or may induce urolith formation in

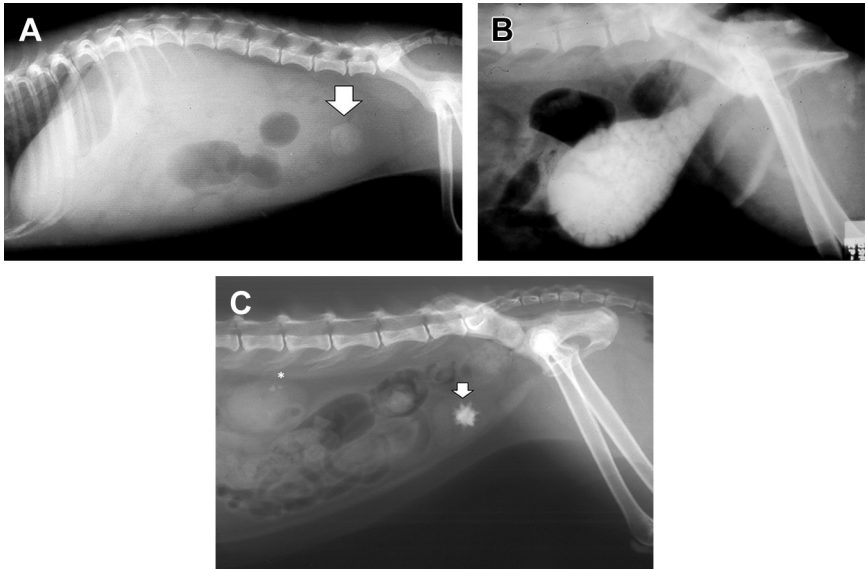


Fig. 1. Radiographic appearance of struvite and calcium oxalate uroliths by abdominal radiography. (A) Lateral abdominal radiograph of a 4-year-old castrated male domestic short-haired cat showing a single round radiopaque sterile struvite urocystolith (*arrow*). (B) Lateral abdominal radiograph of a 3-year-old spayed female Irish setter with numerous variably sized and shaped infection-induced struvite urocystoliths. (C) Lateral abdominal radiograph of an 8-year-old castrated male domestic shorthaired cat showing 1 calcium oxalate dehydrate urocystolith (*arrow*). Renal mineralization is also present (*asterisk*).

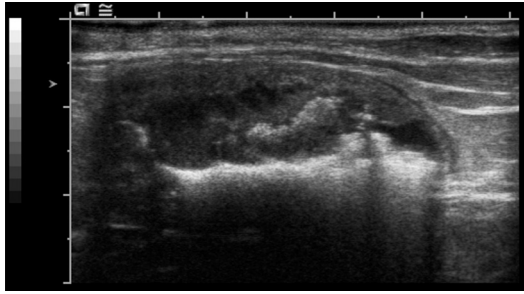


Fig. 2. Images from a 14 year-old, castrated male domestic shorthaired cat with urolithiasis. Urocystoliths appear as shadowing hyperechoic objects on ultrasonographic image of the urinary bladder.

the case of infection-induced struvite uroliths.^{7–9} Factors contributing to this situation include mucosal damage induced by the stones, incomplete urine voiding, or micro-organism entrapment in the stone.

When uroliths are found, it is important to obtain a blood biochemical profile. Blood biochemical results can sometimes suggest presence of underlying diseases such as hypercalcemia that can predispose patients to urolith formation.^{10–13} Because uroliths occasionally cause obstruction, electrolyte, mineral, creatinine, and blood urea nitrogen concentrations should be monitored. Urate calculi may be caused by underlying liver disease, particularly congenital vascular anomalies; therefore, hepatic function should be determined in patients with suspected or confirmed urate uroliths.^{14,15}

DESCRIPTION OF UROLITHS

Determining the composition of uroliths is essential to prevent recurrence. Although many types of uroliths have a characteristic appearance, guessing composition by appearance is unreliable and subject to error.¹⁶ All removed or voided uroliths should be analyzed to determine mineral composition, which aids in developing a successful treatment and prevention plan. Analysis results report the chemical makeup of the different components of the urolith. In cases of recurrence, uroliths should be resubmitted, because mineral composition can change from 1 episode to another.¹⁶

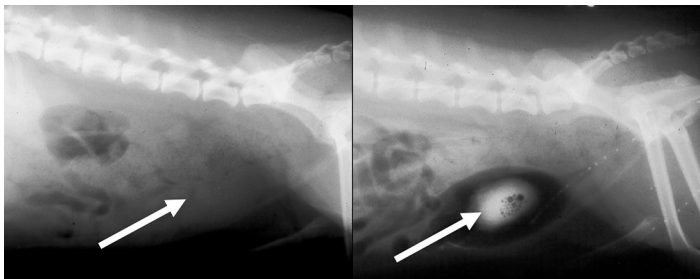


Fig. 3. Survey abdominal radiograph (*left*) and double contrast cystogram (*right*) of a 2 year-old castrated male Pomeranian with radiolucent urate urocystoliths secondary to a portosystemic shunt. The arrows point to the urinary bladder (*left*) and filling defects (urocystoliths, *right*).

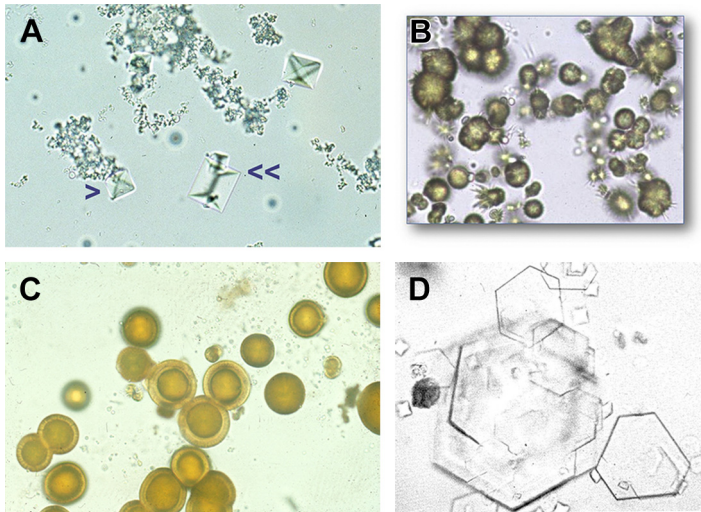


Fig. 4. Crystalluria. (A) Struvite (*double arrowhead*) and calcium oxalate (*single arrowhead*) crystals in a urine sample collected from a 6-year-old castrated male domestic shorthaired cat. (B) Ammonium urate crystals in a urine sample collected from a 1.5-year-old castrated male English bulldog. (C) Xanthine crystals in a urine sample collected from a 3-year-old spayed female beagle administered allopurinol. (D) Cystine crystals in a urine sample collected from a 2-year-old male English bulldog.

A urolith is composed primarily of 1 or more minerals in combination with small quantities of organic matrix. The composition of uroliths may be mixed, with uneven mixtures of minerals throughout the stone or minerals deposited in layers. The different layers of stone are the nidus, stone, shell, and surface crystals. The nidus is the area of obvious initiation of stone growth. The stone refers to the major portion of the urolith. The shell is the material that surrounds the body of the stone and the surface crystals are an incomplete coating of the outer surface of the stone.⁷ A urolith may be defined by a single mineral type, as mixed if the composition consists of more than 1 mineral type, or as a compound stone if there are mineral layers.

MANAGEMENT OF UROCYSTOLITHS

Uroliths may result in clinical signs of lower urinary tract disease, including urethral obstruction. Often, urethral obstruction is associated with azotemia, hyperkalemia, metabolic acidosis, and dehydration.¹⁷ Treatment of urethral obstruction involves relieving the obstruction and correcting the metabolic imbalances as quickly as possible. If urethral obstructions are recurrent, perineal urethrostomy in male cats and scrotal urethrostomy in male dogs may be considered; however, these procedures are associated with increased risk of lower urinary tract disease and bacterial urinary tract infections.¹⁸

Surgical Treatment

Detection of urocystoliths does not necessarily warrant surgical intervention; however, obstruction of urine outflow, an increase in size or number of calculi, persistent clinical signs, and a lack of response to therapy are indications for calculi removal.⁷ Surgery is required in patients with nondissolvable calculi and clinical signs. Traditional open

surgical options are available for treatment of urolithiasis, including cystotomy, urethrotomy, and urethrostomy.

Minimally Invasive Techniques

There are several minimally invasive treatment options for retrieval of bladder and urethral stones. These options include voiding urohydropropulsion, transurethral cystoscopic stone removal with or without use of laser lithotripsy, and minilaparotomy-assisted cystoscopic stone removal, also called percutaneous cystolithotomy (PCCL).

In catheter-assisted retrieval or voiding urohydropropulsion of calculi, the patient is sedated or anesthetized, the bladder is filled with sterile crystalloid solution, and a catheter is passed into the urinary bladder transurethrally.¹⁹ In cats, a 3.5-French or 5-French catheter is used and in dogs a 5-French, 8-French, or 10-French catheter is used, depending on patient size. During catheter retrieval, the contents of the bladder are aspirated while the bladder is agitated by palpating and manipulating it or rotating the patient's body. This procedure is difficult in most male cats and in small male dogs because of the limiting size of the urethra and the size of the catheter. With voiding urohydropropulsion, the patient is held vertically while the distended bladder is manually expressed (**Fig. 5**).¹⁹ Sizes of uroliths that may be retrieved with this technique are approximately 1 mm in male cats, up to 5 mm in female cats, 1 to 3 mm in male dogs, and up to 10 mm in female dogs. These methods are used to eliminate small calculi and to collect them for analysis to plan further treatment. However, these techniques are not successful if a patient presents with urethral obstruction, because this situation indicates that there is at least 1 urolith that is too large to pass through the urethra.

PCCL is a procedure by which the bladder is temporarily fastened to the incised linea, allowing cystoscopic stone removal through a stab incision or a laparoscopic port placed in the urinary bladder (**Fig. 6**).^{20,21} This method is an effective, safe, and efficient means for managing urocystoliths. Cystoscopy produces magnified images of the fluid-distended urinary bladder, allowing identification of abnormalities such as strictures, masses, and calculi.^{22,23} This is the minimally invasive procedure of choice for male dogs and cats, because the diameter of the male urethra limits insertion of a cystoscope with operating channel. Cystoscopic techniques are more

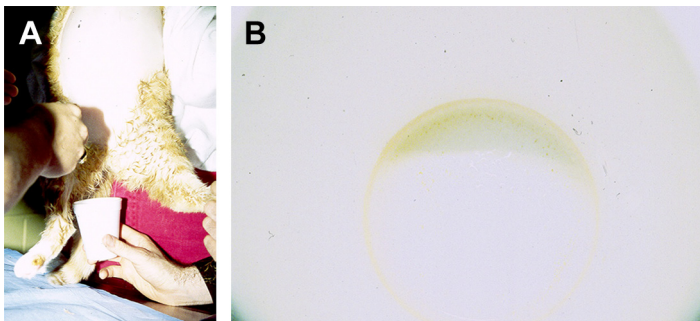


Fig. 5. In voiding urohydropropulsion, the cat is held in a vertical position after the urinary bladder is distended by infusion of fluid through a transurethral catheter. (A) The transurethral catheter is removed and the urinary bladder is gently agitated by grasping it through the abdominal wall. (B) The urinary bladder is gently compressed, inducing micturition and voiding of the urocystoliths.

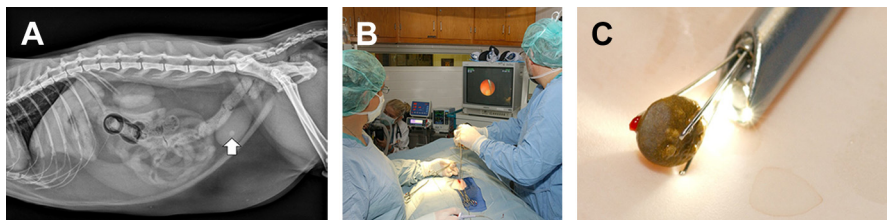


Fig. 6. Minilaparotomy-assisted cystotomy. (A) Lateral abdominal radiograph of an 8-year-old castrated male domestic shorthaired cat with a single urocytolith (arrow). (B) The urinary bladder is grasped through a small incision and tacked to the body wall. A stab incision is made through the bladder wall, and a rigid cystoscope is inserted. Urocytoliths are retrieved using retrieval devices. Cystoscopy provides magnification and better visualization. The urocytolith grasped with 4-prong nitinol graspers is projected on the endoscopic monitor. (C) Urocytolith retrieved via the procedure.

efficient than surgical procedures, decreasing the risk of trauma and abdominal contamination.^{22,23}

In transurethral cystoscopy, a cystoscope is inserted into the urethra and passed into the urinary bladder. This procedure is preferred for use in females but has been described in male dogs because it is less invasive than other diagnostic and treatment methods. If calculi are small enough, they can be removed using stone retrieval devices such as stone baskets and graspers. For larger calculi, lithotripsy may be used if available.²³ Lithotripsy uses a laser fiber, which is passed through the operating channel on the cystoscope. The fiber emits light at an infrared wavelength to fragment calculi,^{24–29} and the resulting fragments are removed transurethrally. This procedure is possible in female dogs and cats and male dogs, but not male cats, because of the limiting size of the male cat urethra and inability to insert a large enough scope with an operating channel.

Medical Management

Struvite

Struvite is another name for crystals or uroliths composed of magnesium ammonium phosphate hexahydrate ($\text{Mg}^{2+}\text{NH}_4^+\text{PO}_4^{3-} \cdot 6\text{H}_2\text{O}$). For uroliths to form, urine must be oversaturated with respect to the minerals that precipitate to form that type of urolith. For struvite uroliths to form, urine must be oversaturated with magnesium, ammonium, and phosphate ions. Urinary oversaturation with struvite may occur as a consequence of a urinary tract infection with a urease-producing microbe (infection-induced struvite) or without the presence of a urinary tract infection (sterile struvite).³⁰ Infection-induced struvite is the most common form occurring in dogs,³¹ whereas sterile struvite is the most common form occurring in cats. Any animal that develops a bacterial urinary tract infection with a urease-producing microorganism can develop infection-induced struvite uroliths. Sterile struvite uroliths have been documented to occur in dogs,³² but they are rare.

Infection-induced struvite

Infection-induced struvite uroliths occur more commonly in dogs and in cats less than 1 year and greater than 10 years of age.^{8,9,33} There is no published information on gender predilection for infection-induced struvite uroliths in cats; however, they occur more commonly in female dogs, because of their increased risk of urinary tract infection. Infection-induced struvite uroliths form because of an infection with a

urease-producing microbe in a fashion similar to human beings.³⁴ In this situation, dietary composition is not important, because the production of the enzyme, urease, by the microbial organism is the driving force behind struvite urolith formation. *Staphylococcus* spp, *Enterococcus* spp, and *Proteus* spp are the most common organisms associated with urease production and infection-induced struvite urolith formation.

Infection-induced struvite uroliths can be dissolved by feeding a struvite dissolution diet and administering an appropriate antimicrobial agent based on bacteriologic culture and sensitivity, although there is a case report³⁵ of dissolution with administration of an antimicrobial agent only. Average dissolution time for infection-induced struvite uroliths was 79 days (range, 64–92 days) in 3 cats reported in a study.³⁰ An alternative dissolution protocol was shown to be effective in 8 of 11 dogs with presumed infection-induced struvite.³⁶ In this protocol, the diet was not changed; instead a urinary acidifier, D,L-methionine, was administered at a dosage of 75 to 100 mg/kg by mouth every 12 hours in combination with an appropriate antibiotic for the organism responsible for struvite formation (typically *Staphylococcus*).³⁶ Median time to dissolution was 2 months, with a range of 1 to 4 months.

It is important that the patient receives an appropriate antimicrobial agent during the entire time of medical dissolution because bacteria become trapped in the matrix of the urolith, and as the urolith dissolves, bacteria are released into urine. If therapeutic levels of an appropriate antimicrobial agent are not present in urine, then an infection will recur, and dissolution will cease.

Prevention of infection-induced struvite does not require feeding a special diet, because the infection causes these struvite uroliths to form. It involves preventing a bacterial urinary tract infection from recurring and treating bacterial infections as they arise. Dietary modification does not prevent infection-induced struvite uroliths from recurring, because diet does not prevent recurrence of a bacterial urinary tract infection.

Sterile struvite

Sterile struvite uroliths form typically in cats between 1 and 10 years of age.³³ Risk for struvite urolith formation decreases after approximately 6 to 8 years of age in cats.³⁷ Struvite uroliths occur with equal frequency in male and female cats. Sterile struvite uroliths form because of dietary composition as well as innate risks for urolith formation. Experimentally, magnesium phosphate and struvite uroliths formed in healthy cats consuming calculogenic diets containing 0.15% to 1.0% magnesium (dry matter basis).^{34,38,39} However, these data are difficult to interpret, because the amount of magnesium consumption by cats in these studies may be different from that by cats that spontaneously form sterile struvite uroliths consuming commercial diets, because of differences in caloric density, palatability, and digestibility.⁴⁰ The influence of magnesium on struvite formation depends on urine pH⁴¹ and influence of ions, minerals, and other components in urine.⁴² Alkaluria is associated with increased risk for struvite formation.^{43,44} In a clinical study including 20 cats with naturally occurring struvite urocystoliths and no detectable bacterial urinary tract infection,³⁰ the mean urinary pH at the time of diagnosis was 6.9 ± 0.4 . An additional factor is water intake and urine volume. Consumption of increased quantities of water may result in decreasing concentrations of calculogenic substances in urine, thus decreasing risk of urolith formation.⁴⁵ Consumption of small quantities of food frequently rather than 1 or 2 large meals per day is associated with production of more acidic urine and a lesser degree of struvite crystalluria by cats.^{46,47}

Sterile struvite uroliths can be dissolved by feeding a diet that is restricted in magnesium, phosphorous, and protein and that induces aciduria relative to maintenance

adult cat foods.^{30,48} In a clinical study including 22 cats with sterile struvite urocystoliths,³⁰ urocystoliths dissolved in 20 cats in a mean of 36.2 ± 26.6 days (range, 14–141 days). The cats were fed a high-moisture (canned), calorically dense diet containing 0.058% magnesium (dry matter basis) and increased sodium chloride (0.79% dry matter basis). Similar results have been found in another study.⁴⁹ In a more recent study of 32 cats with presumed struvite urocystoliths comparing 2 low magnesium acidifying diets, one a struvite dissolution diet and the other a struvite prevention diet, the mean ($\pm$ standard deviation [SD]) times for a 50% reduction in urolith size (0.69 ± 0.1 weeks) and complete urolith dissolution (13.0 ± 2.6 days) were significantly shorter for cats fed the struvite dissolution diet, compared with those (1.75 ± 0.27 weeks and 27.0 ± 2.6 days, respectively) for cats fed the struvite prevention diet. At study termination, mean $\pm$ SD urine pH (6.083 ± 0.105) for cats fed the struvite dissolution diet was lower than that (6.431 ± 0.109) for cats fed the struvite prevention diet.⁵⁰ Therefore, sterile struvite urocystoliths often dissolve in 2 to 4 weeks.

Prevention of sterile struvite uroliths involves inducing a urine pH less than approximately 6.8,^{41,51} increasing urine volume, and decreasing excretion of magnesium, ammonium, and phosphorous. There are many diets available that are formulated to be struvite preventive.³³

Calcium oxalate

Calcium oxalate accounts for 40% to 50% of all uroliths. Risk factors for calcium oxalate formation include increased urinary or oxalate excretion and aciduria. Certain breeds of dogs and cats are predisposed to formation of calcium oxalate uroliths, including long-haired cats (Burmese, Persian, and Himalayan breeds) and small breeds dogs (miniature schnauzer, Lhasa apso, shih tzu, and Yorkshire terrier breeds).^{52–54} Calcium oxalate urolith formation occurs when urine is oversaturated with calcium and oxalate.¹ In addition to these alterations in activities of ions, large-molecular-weight proteins occurring in urine, such as nephrocalcin, uropontin, and Tamm-Horsfall mucoprotein, influence calcium oxalate formation.⁵⁵ We have a limited understanding of the role of these macromolecular and ionic inhibitors of calcium oxalate formation in cats. Certain metabolic factors are known to increase risk of calcium oxalate urolith formation in several species, including cats and dogs. Medical and nutritional strategies for stone prevention have focused on amelioration of these factors.

Calcium homeostasis is achieved through actions of parathyroid hormone (PTH) and 1,25-dihydroxycholecalciferol (1,25-vitamin D) on bones, intestines, and kidneys. When serum ionized calcium concentration decreases, PTH and 1,25-vitamin D activities increase, resulting in mobilization of calcium from bone, increased absorption of calcium from intestine, and increased reabsorption of calcium by renal tubules. Conversely, high serum ionized calcium concentration suppresses release of PTH and production of 1,25-vitamin D, resulting in decreased bone mobilization, decreased intestinal absorption of calcium, and increased urinary excretion of calcium. Therefore, hypercalciuria can result from hypercalcemia, excessive intestinal absorption of calcium (gastrointestinal [GI] hyperabsorption), impaired renal reabsorption of calcium (renal leak), or excessive skeletal mobilization of calcium (resorptive).

Hypercalcemia is associated with increased risk of calcium oxalate urolith formation. In cats with calcium oxalate uroliths, hypercalcemia was observed in 35% of the cases,⁵⁶ whereas in dogs with calcium oxalate uroliths, it occurs in approximately 4%, usually associated with primary hyperparathyroidism.⁵⁷ Conversely, uroliths developed in 35% of cats with idiopathic hypercalcemia.¹² Hypercalcemia results in increased calcium fractional excretion and hypercalciuria when severe.

Hypercalciuria is a significant risk factor, but not necessarily the cause of calcium oxalate urolith formation in human beings, dogs, and cats.⁵⁸ In miniature schnauzers, GI hyperabsorption seems to occur most commonly, although renal leak hypercalciuria has also been observed.⁵⁹ Hypercalciuria has not been well defined in normocalcemic cats with calcium oxalate uroliths but is believed to occur. Although excessive dietary intake of calcium may result in hypercalciuria, studies in human beings refute this. Apparently, dietary calcium may bind to dietary oxalic acid, resulting in calcium oxalate formation in the lumen of the GI tract, thereby preventing absorption of calcium and oxalate. Hypercalciuria may also occur with administration of loop diuretics, glucocorticoids, urinary acidifiers, and vitamin D or C.

Metabolic acidosis promotes hypercalciuria by promoting bone turnover (release of calcium with buffers from bone), increasing serum ionized calcium concentration, resulting in increased urinary calcium excretion and decreased renal tubular reabsorption of calcium. Consumption of diets supplemented with the urinary acidifier ammonium chloride by cats has been associated with increased urinary calcium excretion.⁶⁰ In addition, consumption by human beings of diets containing high amounts of animal protein results in metabolic acidosis and increased urinary calcium excretion. Metabolic acidosis promotes hypercalciuria by promoting bone turnover (release of calcium with buffers from bone), increasing serum ionized calcium concentration, resulting in increased urinary calcium excretion and decreased renal tubular reabsorption of calcium. In addition, acidic urine alters function and concentration of crystal inhibitors. Low urine pH decreases urinary citrate concentration by increasing renal proximal tubular citrate reabsorption. Acidic urine is known to impair function of macromolecular protein inhibitors. In dogs, hypercalciuria resulting from ammonium chloride administration was decreased by bicarbonate administration.⁶¹ In cats, magnesium supplementation as magnesium chloride was associated with increased urinary calcium excretion and aciduria, whereas magnesium supplementation as magnesium oxide was associated with alkaluria and a lesser degree of urinary calcium excretion.⁴¹ Urine pH has a direct effect on solubility of calcium oxalate. In a study of healthy cats fed similar diets that differed in only their acidifying or alkalinizing properties, urinary saturation with calcium oxalate was lower when the urine pH was greater than 7.2 and greater when urine pH was less than 6.5.⁵¹

Inhibitors, such as citrate, magnesium, and pyrophosphate, form soluble salts with calcium or oxalic acid and reduce availability of calcium or oxalic acid for precipitation. Other inhibitors, such as Tamm-Horsfall glycoprotein and nephrocalcin, interfere with the ability of calcium and oxalic acid to combine minimizing crystal formation, aggregation, and growth.

Oxalic acid is a metabolic end product of ascorbic acid (vitamin C) and several amino acids, such as glycine and serine, derived from dietary sources. Oxalic acid forms soluble salts with sodium and potassium ions, but a relatively insoluble salt with calcium ions. Therefore, any increased urinary concentration of oxalic acid may promote calcium oxalate formation. Dietary increases of oxalate and vitamin B₆ deficiency are known factors increasing urinary oxalate. Hyperoxaluria has been observed experimentally in kittens consuming vitamin B₆-deficient diets¹¹ but has not been associated with formation of naturally occurring calcium oxalate uroliths in adults. Genetic anomalies may also increase urine oxalic acid concentration. Hyperoxaluria has also been recognized in a group of related cats with reduced quantities of hepatic D-glycerate dehydrogenase, an enzyme involved in metabolism of oxalic acid precursors (primary hyperoxaluria type II).⁶² Hyperoxaluria has also been associated with defective peroxisomal alanine/glyoxylate aminotransferase activity (primary hyperoxaluria type I) and intestinal disease in human beings (enteric hyperoxaluria). These

conditions have not been evaluated in cats or dogs. *Oxalobacter formigenes* is an enteric bacterial organism that metabolizes oxalic acid in the GI tract. It has recently been shown that dogs that have less enteric colonization with *Oxalobacter formigenes* have a higher risk of calcium oxalate urolith formation than dogs that are colonized.⁶³ Compared with the situation in human beings, urinary oxalate seems to have a lesser role in calcium oxalate formation in dogs and cats and urinary calcium seems to have a greater role.^{64,65}

Decreased urine volume results in increased calcium and oxalic acid saturation and an increased risk for urolith formation. Cats can achieve urine specific gravities in excess of 1.065, indicating a marked ability to produce concentrated urine. Many cats affected with calcium oxalate uroliths have a urine specific gravity greater than 1.040 unless there is some impairment of renal function or concentrating ability.⁵⁸ Detection of calcium oxalate crystals indicates that urine is supersaturated with calcium oxalate and, if persistent, represents an increased risk for calcium oxalate urolith formation. However, calcium oxalate crystalluria is present in less than 50% of feline and canine cases at time of diagnosis of urolithiasis.⁵⁸

Medical protocols that promote dissolution of calcium oxalate uroliths are not available; therefore, uroliths must be removed physically, either surgically or by voiding urohydropulsion.⁶⁶

Calcium oxalate uroliths are recurrent; therefore, preventive measures are warranted. There is an approximate 10% recurrence at 6 months and 35% recurrence at 12 months in dogs,⁷ and many cats have recurrence within 2 years of removal if preventive measures are not undertaken.⁶⁷ Because the cause of calcium oxalate urolith formation is not completely known, no treatment has been shown to be completely effective. If possible, metabolic factors known to increase calcium oxalate risk should be corrected or minimized. Goals of dietary prevention include (1) reducing urine calcium and oxalate concentration, (2) promoting high concentrations and activity of urolith inhibitors, (3) reducing urine acidity, and (4) promoting dilute urine.

Increasing urine volume is a mainstay of preventive therapy for calcium oxalate urolithiasis. By increasing water intake, urinary concentrations of calculogenic minerals are reduced. In addition, larger urine volumes typically increase urine transit time and voiding frequency, thereby reducing retention time for crystal formation and growth. Feeding a canned food is the most practical means of increasing water intake and decreasing calcium oxalate urine saturation. The goal is to dilute urine to a specific gravity of 1.040 or less in cats and 1.030 or less in dogs.^{7,68} Flavoring water, enhancing water access, and adding water to dry foods may be used in patients that refuse to eat canned foods. Sodium chloride may be used to increase water intake, and several calcium oxalate preventive diets contain greater than 1% sodium chloride (dry matter basis).^{69,70} However, increased dietary sodium may increase urinary calcium excretion and can contribute to ongoing renal damage in cats with marginal renal function,⁶⁸ although this has not been a consistent finding.^{71,72}

Epidemiologic studies consistently identify acidifying diets among the most prominent risk factors for calcium oxalate urolithiasis.^{52,53,73} Solubility of calcium oxalate in urine is minimally influenced by pH; however, there is a linear relationship between increasing calcium oxalate solubility and urine pH in healthy cats, with alkaluria inducing a lower relative supersaturation for calcium oxalate when compared with aciduria.⁵¹ Persistent aciduria may be associated with low-grade metabolic acidosis, which promotes bone mobilization and increases urinary calcium excretion; however, this effect has not been reported in studies of cats for up to 1 year.⁵¹ In a case series of 5 cats with hypercalcemia and calcium oxalate uroliths,⁵³ discontinuation of acidifying diets or urinary acidifiers was associated with normalization of serum calcium

concentration. In dogs, the influence of urine pH on risk of calcium oxalate formation is less clear.⁷⁴ Furthermore, aciduria promotes hypocitraturia and functional impairment of endogenous urolith inhibitors. Thus, feeding an acidifying diet or administering urinary acidifiers to cats at risk for calcium oxalate is contraindicated. A target urine pH of 6.6 to 7.5 is suggested in dogs and cats at risk for recurrence of calcium oxalate uroliths.⁶⁸

Potassium citrate is often included in diets designed for calcium oxalate prevention. In urine, citric acid combines with calcium to form soluble complexes, thereby reducing ionic calcium concentration. Citric acid also directly inhibits nucleation of calcium and oxalate crystals. When oxidized within the tricarboxylic acid cycle, supplemental citrate results in urine alkalization caused by production of bicarbonate. The metabolic alkalization increases endogenous renal citrate excretion and reduces calcium absorption and urinary excretion.⁶⁸ Commercial products that add citrate but continue to acidify the urine (pH <6.5) negate the benefit of citrate therapy.

Although reduction of urine calcium and oxalic acid concentrations by restriction of dietary calcium and oxalic acid seems logical, it is not without risk. Reducing consumption of only one of these constituents may increase availability and intestinal absorption of the other, resulting in increased urinary excretion. Conversely, increasing dietary calcium levels in normal cats contributes directly to increased urine calcium concentration. Because epidemiologic data in cats suggest that marked dietary calcium restriction increases urolith risk, moderate levels of dietary calcium are advised in nonhypercalcemic cats.⁶⁸

Dietary phosphorus should not be restricted in dogs or cats with calcium oxalate urolithiasis. Low dietary phosphorus is a risk factor for calcium oxalate urolith formation in cats.⁷⁵ Reduction in dietary phosphorus may be associated with activation of vitamin D, which in turn promotes intestinal calcium absorption and hypercalciuria. In addition, phosphate status determines pyrophosphate urinary concentrations, an inhibitor of calcium oxalate urolith formation in human beings and rodents. If calcium oxalate urolithiasis is associated with hypophosphatemia and normal calcium concentration, oral phosphorus supplementation may be considered. However, caution should be used because excessive dietary phosphorus may predispose to formation of calcium phosphate uroliths. Whether this occurs in dogs or cats is unknown.

Urinary magnesium forms complexes with oxalic acid, reducing the amount of oxalic acid available to form calcium oxalate. Studies in cats associate low dietary magnesium with calcium oxalate risk.^{37,53,73,75–77} In human beings, supplemental magnesium has been used to minimize recurrence of calcium oxalate uroliths; however, supplemental magnesium may increase the risk of struvite formation in cats. Risks and benefits of magnesium supplementation to dogs and cats with calcium oxalate urolithiasis have not been evaluated and it is not advised. It seems logical that magnesium should not be highly restricted in diets that are consumed by dogs or cats with calcium oxalate urolithiasis.

Consumption of high amounts of animal protein by human beings is associated with an increased risk of calcium oxalate formation. Dietary protein of animal origin may increase urinary calcium and oxalic acid excretion, decrease urinary citrate excretion, and promote bone mobilization to buffer the acid intake from metabolism of animal proteins. However, a case control study⁷⁵ showed that higher protein concentration in cat foods appeared protective against calcium oxalate uroliths. Although several coassociations (eg, higher protein in canned foods) might explain this finding, cats are obligatory carnivores and dietary protein restriction in the management of calcium oxalate urolithiasis is not advised.

Excess intake of vitamin C, a metabolic oxalate precursor, should be avoided.⁶⁸ Although normal dietary vitamin C levels are not considered a risk in human beings, very small increases in urinary oxalate are a concern in urolith formers. Because dogs and cats do not have a dietary vitamin C requirement, supplementation should be avoided in foods fed to dogs or cats at risk for calcium oxalate uroliths. Cranberry concentrate tablets are also contraindicated. They provide mild acidification and are high in oxalate, as well as vitamin C.⁷⁸

The diet should be adequately fortified with vitamin B₆, because vitamin B₆ deficiency promotes endogenous production and subsequent urinary excretion of oxalic acid.⁷⁹ There is no evidence that providing increased vitamin B₆ beyond meeting the nutritional requirement provides a benefit in cats. Because most commercial diets designed for dogs and cats are well fortified with vitamin B₆, it is unlikely additional supplementation is beneficial except in homemade diets. Regardless, vitamin B₆ is reasonably safe and sometimes provided to patients with persistent calcium oxalate crystalluria or frequent recurrences.

Increased dietary fiber intake is associated with decreasing risk of calcium oxalate recurrence in some human beings but not in cats unless they are hypercalcemic. Certain types of fiber (soy or rice bran) decrease calcium absorption from the GI tract, which may decrease urinary calcium excretion. Also, higher-fiber diets tend to be less acidifying. In 5 cats with idiopathic hypercalcemia and calcium oxalate uroliths, feeding a high-fiber diet with supplemental potassium citrate resulted in normalization of serum calcium concentrations¹¹; however, efficacy of increased fiber intake is unproved. There are no studies in dogs evaluating dietary fiber and calcium oxalate urolithiasis.

In dogs, feeding a protein-restricted and sodium-restricted alkalinizing diet has been shown to decrease recurrence of calcium oxalate uroliths.⁷⁷ If a neutral to slightly alkaline urine pH is not accomplished by diet, potassium citrate may be administered (initial dose: 75 mg/kg by mouth every 12 hours; adjust to induce a urine pH of 7.0–7.5). Because protein-restricted and sodium-restricted alkalinizing diet is also high in fat, some dogs cannot tolerate the diet. In these dogs, feeding a low-fat, higher-fiber diet with supplemental potassium citrate seems to be effective.

In cats, several diets are available that are formulated to reduce calcium and oxalic acid concentrations in urine, promote high concentration and activity of inhibitors of calcium oxalate crystal growth and aggregation in urine, and maintain dilute urine. Consumption of these diets by healthy cats results in production of urine that is undersaturated with calcium oxalate.^{37,45} In 1 study of cats with naturally occurring calcium oxalate uroliths,⁶⁷ consumption of 1 oxalate preventive diet resulted in a decrease of urine saturation from the oversaturated state to a metastable state; cats did not reform uroliths. In cats with hypercalcemia and calcium oxalate uroliths, prevention seems to be more successful when feeding a higher-fiber diet, and administering potassium citrate (initial dose: 75 mg/kg by mouth every 12 hours; adjust to induce a urine pH of 7.0–7.5).¹¹ Other treatments that have been proposed include vitamin B₆ (2 mg/kg by mouth every 24 hours) and hydrochlorothiazide (2–4 mg/kg by mouth every 12 hours).

Calcium oxalate uroliths are recurrent. Serial monitoring of a dog or cat with a history of calcium oxalate urolithiasis should be part of the preventive protocol. Periodically, a complete urinalysis should be performed to monitor urine specific gravity, pH, and presence of calcium oxalate crystalluria. Ideally, urine should be dilute, urine pH in the neutral to alkaline range, and calcium oxalate crystalluria should not be present. Survey abdominal radiography should be performed approximately every 6 months to evaluate for recurrence. If calcium oxalate urocystoliths are detected while small,

they may be retrieved nonsurgically and adjustment to the preventive protocol can be made.

Purines

Urates Most information concerning urate uroliths is derived from dogs, with little information available for cats. Uric acid is one of several biodegradation products of purine nucleotide metabolism.⁸⁰ Purines are made up of 3 groups of compounds: oxypurines (hypoxanthine, xanthine, uric acid, allantoin), aminopurines (adenine, guanine), and methylpurines (caffeine, theophylline, theobromine). In most dogs and cats, allantoin is the major metabolic end product; it is the most soluble of the purine metabolic products excreted in urine. Uroliths that are composed of uric acid (anhydrous uric acid, uric acid dihydrate, sodium urate, ammonium urate) or xanthine form because urine is oversaturated with these substances. Ammonium urate (also known as ammonium acid urate and ammonium biurate) is the monobasic ammonium salt of uric acid. It is the most common form of naturally occurring purine uroliths observed in dogs and cats. Other naturally occurring purine uroliths include sodium urate (also known as sodium acid urate or monosodium urate), sodium calcium urate, potassium urate, and uric acid dihydrate.

Urate is the third most common mineral found in uroliths in dogs and cats, accounting for 5% to 8% of uroliths, and the second most common urolith, occurring in dogs and cats less than 1 year of age (infection-induced struvite is the most common urolith in these patients). Urate uroliths form when urine is oversaturated with urate and usually ammonium. These uroliths form because of liver disease (usually a portosystemic vascular shunt)^{14,15} or because of an inborn error of metabolism resulting in hyperuricosuria (eg, Dalmatians and English bulldogs).^{81,82} They are more common in dogs and cats less than 7 years of age.^{81–83}

Dalmatian dogs are predisposed to urate uroliths because of the unique metabolism of purines.^{81,84} The ability of Dalmatians to oxidize uric acid to allantoin is intermediate between human beings and most non-Dalmatian dogs.⁸⁵ Human beings have a serum uric acid concentration of approximately 3 to 7 mg/dL, and excrete approximately 500 to 700 mg of uric acid in their urine per day.⁸⁶ Most non-Dalmatian dogs have a serum uric acid concentration of less than 0.5 mg/dL and excrete approximately 10 to 60 mg of uric acid in their urine per day. Dalmatians have a serum uric acid concentration that is 2 to 4 times that of non-Dalmatians and excrete more than 400 to 600 mg of uric acid in their urine per day.⁸⁰

Studies of the fate of uric acid in Dalmatians have shown unique hepatic and renal pathways of metabolism.^{84,87} Of these 2 metabolic sites, reciprocal allogenic renal and hepatic transplantations between Dalmatians and non-Dalmatians indicate that the hepatic mechanism is quantitatively the most significant. The liver of Dalmatians does not completely oxidize available uric acid, even although it contains a sufficient concentration of uricase.⁸⁸ It has been shown that Dalmatians, English bulldogs, and black Russian terriers have a mutation in the SLC2A9 gene that encodes for a transporter of uric acid, and homozygous mutation results in lack of hepatic conversion of uric acid to allantoin.^{89,90} The definitive mechanism of urate urolith formation in Dalmatian dogs remains unknown. Increased uric acid excretion is a risk factor rather than a primary cause. Urate uroliths are reported more commonly in males than females; the average age of urolith diagnosis is 4.5 years.^{80,81} Although all Dalmatians excreting high uric acid excrete relatively increased quantities of uric acid in their urine, apparently, only a small percentage form urate stones.⁹¹

Dissolution is not possible in dogs and cats with uncorrected liver disease (eg, nonsurgical portovascular anomalies or microvascular dysplasia). Surgical removal,

voiding urohydropulsion, or cystoscopy $\pm$ laser lithotripsy remains the treatment of choice for symptomatic urate stones that cannot be dissolved. It is logical to hypothesize that elimination of hyperuric aciduria and reduction of urine ammonium concentration after surgical correction of anomalous shunts would result in spontaneous dissolution of uroliths composed primarily of ammonium urate. Appropriate clinical studies are needed to prove or disprove this hypothesis. We have occasionally been successful in medically dissolving urate uroliths in dogs with portal vascular anomalies but have not attempted dissolution in cats with ammonium urate uroliths and portal vascular anomalies. Additional clinical studies are needed to evaluate the relative value of calculolytic diets, allopurinol, or alkalinization of urine in dissolving ammonium urate uroliths in cats with portal vascular anomalies. The pharmacokinetics and efficacy of allopurinol may be altered in cats with portal vascular anomalies, because biotransformation of this drug, which has a very short half-life, to oxypurinol, which has a longer half-life, requires adequate hepatic function. Xanthine uroliths have been observed to form in dogs with portovascular anomalies given allopurinol; therefore, allopurinol had an effect on xanthine oxidase conversion of xanthine to uric acid.

In dogs and cats without underlying liver disease, dissolution may be attempted. Dissolution of urate uroliths in dogs is accomplished by feeding a purine-restricted, alkalinizing, diuresing diet and administering the xanthine oxidase inhibitor, allopurinol (15 mg/kg by mouth every 12 hours)^{80,92–96}; allopurinol has not been evaluated in cats and should not be used. Although renal failure diets are protein restricted and thus lower in purines, there are 2 commercially available diets formulated to be low in purines: Prescription Diet u/d (Hill's Pet Products) and UC Low Purine (Royal Canin). Canned diets may be better than dry diets. In 1 study,⁸⁰ medical dissolution was effective in approximately 40% of Dalmatians, partial dissolution occurred in approximately 30%, and no dissolution of growth of uroliths caused by xanthine formation occurred in approximately 30%. Although no studies have been performed evaluating the efficacy or safety of medical dissolution of urate uroliths in cats with idiopathic urate urolithiasis, we have successfully dissolved urate uroliths in cats using a renal failure diet (Prescription Diet k/d, Hill's Pet Products) and allopurinol (7.5 mg/kg by mouth every 12 hours). Until further studies are performed to confirm the safety and efficacy of medical dissolution, surgical removal remains the treatment of choice for urate uroliths in cats.

Prevention of urate uroliths in dogs without liver disease involves continued feeding of the low purine diet. If necessary, allopurinol may be administered at a lower dose (7–10 mg/kg by mouth every 12–24 hours). Periodic abdominal ultrasonography or double contrast cystography may be necessary. Determination of 24-hour urinary uric acid excretion may also be useful, although data are limited and collection of 24-hour urine samples can be difficult. The goal is to achieve 24-hour urinary acid excretion of approximately 250 to 350 mg/dog/24 h. Use of urine uric acid/urine creatinine ratios may be useful for diagnosis of risk of urate urolith formation⁹⁰ but are inaccurate for predicting success of dissolution or prevention.⁸⁰ Prevention of urate urolith recurrence in cats has been greater than 90% when using a protein-restricted alkalinizing diet.

Xanthine Xanthine urolithiasis may occur with allopurinol administration to dogs, especially when dietary purines are not restricted. Management involves adjusting dosage of allopurinol and changing diet. Spontaneously occurring xanthine uroliths have been reported rarely in cats (0.14% of uroliths analyzed at the Minnesota Urolith Center). They are composed typically of pure xanthine, and they have been reported to occur in cats less than 5 years of age, with approximately even distribution between

males and females. Often, xanthine uroliths occur in multiple numbers but are smaller than 5 mm; they are radiolucent. Of 64 cats that formed xanthine uroliths in 1 report, none of the cats had been treated with the xanthine oxidase inhibitor allopurinol. Sixty-one xanthine uroliths were obtained from the lower urinary tract, whereas xanthine uroliths from 3 cats came from the upper urinary tract. Xanthine uroliths occurred in 30 neutered and 8 nonneutered males and 25 neutered females (the gender of 1 cat was not specified). Mean age of cats at time of diagnosis of xanthine uroliths was 2.8 ± 2.3 years (range, 4 months to 10 years). Eight of the 64 cats were less than 1 year old. Urinary uric acid excretion was similar between 8 xanthine urolith-forming cats and healthy cats (2.09 ± 0.8 mg/kg/d vs 1.46 ± 0.56 mg/kg/d); however, urinary xanthine excretion (2.46 ± 1.17 mg/kg/d) and urinary hypoxanthine excretion (0.65 ± 0.17 mg/kg/d) were higher (neither is detectable in urine from healthy cats). Xanthine uroliths have also been found in a few Cavalier King Charles spaniels.^{97–99} No medical dissolution protocol for naturally occurring xanthine uroliths exists. Prevention involves feeding a purine-restricted, alkalinizing, diuresing diet. Without preventive measures, xanthine uroliths often recur within 3 to 12 months after removal.

No medical dissolution protocol for feline xanthine uroliths exists. Prevention involves feeding a protein-restricted alkalinizing diet. Without preventive measures, xanthine uroliths often recur within 3 to 12 months after removal. In an unpublished observational study (JW Bartges, personal communication, 2014), 10 cats with confirmation of xanthine uroliths by quantitative analysis had been consuming the protein-restricted alkalinizing diet and followed for at least 2 years; only one had had recurrence.

Cystine

Cystinuria occurs when there is a proximal renal tubular defect in reabsorption and often occurs with other amino acids (notably, ornithine, lysine, and arginine).^{100–105} Affected animals also show altered intestinal transport of cystine.^{106–108} Several genetic mutations have been identified in dogs and cats that are associated with cystinuria.^{109–112} Cystinuria is associated only with urolith formation and is not associated with protein malnutrition or amino acid deficiency, although it can be associated with hypercarnitinuria or hypertaurinuria and associated dilated cardiomyopathy.¹¹³ However, cystinuria by itself does not result in urolithiasis, and many cystinuric dogs and human beings do not form uroliths.¹¹⁴

Canine cystine uroliths can be dissolved medically. Feed a diet that is low protein, alkalinizing, and induces a diuresis. Although renal failure diets are protein restricted and thus lower in amino acids, there are 2 commercially available diets formulated to be low in sulfur-containing amino acids: Prescription Diet u/d (Hill's Pet Products) and UC Low Purine (Royal Canin). Canned diets may be better than dry diets. Administer 2-mercaptopyropionylglycine (2-MPG, Tiopronin, Thiola).^{100,101,115} This drug is similar to D-penicillamine in that it binds to the individual cysteine molecules, preventing formation of the disulfide bond of cystine; however, it is associated with fewer side effects and complications than D-penicillamine. Dosage is 15 mg/kg by mouth every 12 hours for dissolution. There are studies showing that administration of 2-MPG without modifying diet may result in dissolution of cystine uroliths. Cats do not tolerate 2-MPG well and it is associated with GI signs, liver disease, and anemia; therefore, it should not be used in cats; dissolution of cystine uroliths has not been successful.

Prevention of cystine uroliths involves inducing undersaturation of urine with cystine. Continued feeding of the low sulfur-containing amino acid diets to dogs and feeding a renal failure diet to cats has been successful in prevention of cystine

urolith recurrence. Cystine solubility in urine is dependent on pH and the solubility increases exponentially when the urine pH is greater than 7.2; therefore, maintain an alkaline urine pH. If additional alkalinization is required, potassium citrate may be administered (initial dosage: 75 mg/kg by mouth every 12 hours; adjust to urine pH >7.2). 2-MPG may be used for prevention; however, many dogs do not require it and, as mentioned, it should not be used in cats.

Mixed and compound uroliths

Between 5% and 15% of uroliths may be mixed or compound stones. This observation refers to a situation in which more than 1 mineral is present within the stone, either mixed within all layers or different parts of the stone are composed of different minerals (nidus vs major volume of the stone vs the outer layer or shell). As mentioned earlier, ammonium urate and calcium carbonate (or calcium apatite) may be mixed with struvite as part of infection-induced struvite stone formation. In this situation, the struvite caused by the infection is the primary focus for prevention. Some stones may be composed of both struvite and calcium oxalate. In this situation, usually calcium oxalate is the nidus (inner part of the stone), and struvite is layered around it. This compound stone forms because the patient first formed a calcium oxalate stone and then developed a urinary tract infection and layered infection-induced struvite around the calcium oxalate nidus. In this situation, preventive therapy is directed toward the inner calcium oxalate component, because calcium oxalate formed first and this led to infection-induced struvite formation. Management of infection is also important; however, dietary prevention is focused on the calcium oxalate component. Ammonium urate uroliths may contain xanthine in patients that receive allopurinol for dissolution or prevention. This situation may be because of (1) too high a dose of allopurinol, (2) lack of restriction of dietary protein/purine, or (3) individual patient metabolism of allopurinol despite appropriate allopurinol dosage and dietary management. Prevention is directed to (1) decreasing or discontinuing allopurinol, (2) changing the diet to a more protein-/purine-restricted diet, or (3) both.

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